INDIA-FIRST PRODUCT, BUSINESS AND BACKEND BLUEPRINT

School Organization B2B SaaS

A multi-tenant learning operations platform sold to schools and education organizations - not to individual students or teachers.

PRODUCT
Problem, users, scope and defensible position.

BUSINESS
Market, school sales, pricing and metrics.

PLATFORM
Tenancy, configuration, data and security.

DELIVERY
MVP, roadmap, risks and operating team.

Decision document | Version 1.0 | 6 September 2026

01 / PRODUCT THESIS

Executive decision

The school or education organization is the paying customer and tenant. Teachers, students, parents and staff are provisioned users inside that tenant.

The one-sentence product

A school learning-operations system that gives management control, gives teachers a fast daily workflow, and gives students and parents one trusted academic timeline.

Who pays

School owner, principal, trust, chain or education organization. Contract and invoice are organization-level.

Who administers

School admins configure academics, rosters, permissions, reports and workflows inside the tenant.

Who uses

Teachers, students, parents, coordinators and staff. They do not buy separate subscriptions.

What renews

Verified adoption, saved staff time, visible student follow-up and dependable support.

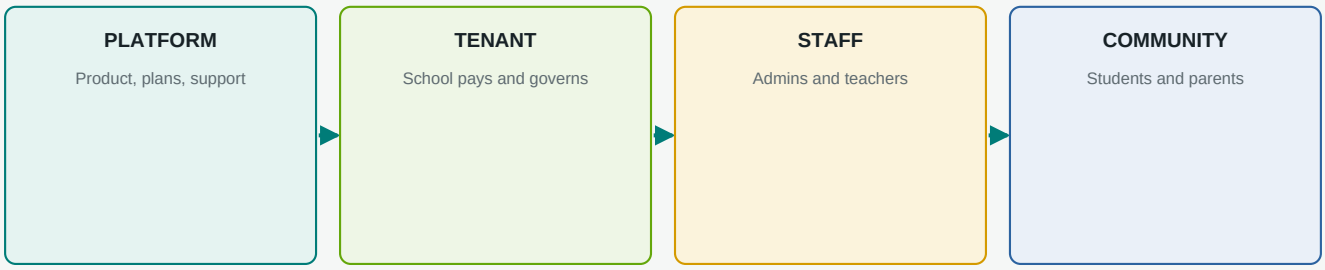
Strategic choices

- Start with the daily academic loop: roster -> class -> attendance -> assignment -> submission -> grade -> intervention.
- Build a modular monolith in NestJS with PostgreSQL as the system of record; split services only after measured pressure.
- Use configuration and entitlements, not per-school code forks. Protect data, permissions and audit invariants.
- Sell a paid pilot, convert to an annual contract, and make onboarding and teacher adoption part of the product.
- Defer full ERP scope, MongoDB, independent AI services, Kubernetes and generalized workflow engines.

02 / COMMERCIAL MODEL

Customer and tenant boundaries

The platform company operates the service. The school governs its tenant. People use only the roles and data their school has authorized.



Platform super admin may

Provision or suspend tenants; manage plans and entitlements; observe service health; use audited support access; manage billing, incidents and aggregate usage.

Platform super admin must not

Silently change grades or attendance; browse student content without a support purpose; mix school data; bypass policy; use child data for model training without lawful agreement.

Commercial boundaries

- One legal customer can own one tenant or a multi-campus tenant. A school chain can receive a group view while campuses keep delegated control.
- Subscriptions, invoices, SLAs, storage limits and AI budgets attach to the tenant account.
- A user may belong to multiple organizations, but every session selects one tenant and every request is tenant-scoped.
- Contracts must define data roles, retention, subprocessors, incident handling, exports and termination support.

03 / BUYER PAIN

Problems worth solving

Schools buy control, adoption, evidence and lower operating friction. Feature count alone does not create a reason to switch.

PROBLEM	CURRENT FAILURE	OUTCOME SOLD
Fragmented work	Attendance, homework, marks, files and messages live in separate tools.	One daily academic workflow and timeline.
Low visibility	Management sees reports late and cannot spot classes or students falling behind.	Live adoption, completion and intervention views.
Teacher admin load	Repeated entry, reminders and spreadsheet consolidation consume teaching time.	Fast entry, bulk actions, templates and automation.
Parent uncertainty	Families receive irregular updates and chase staff for status.	Policy-controlled summaries and notifications.
Weak accountability	Changes to marks, attendance and access are hard to reconstruct.	Audited records, approvals and role boundaries.
Vendor lock-in	Custom school software becomes expensive to upgrade or integrate.	Configurable SaaS plus standard imports, exports and APIs.

North-star outcome

By Friday afternoon, a principal should know which classes operated, which teachers completed core workflows, which students need attention, and whether the system is being used - without requesting a spreadsheet.

04 / PRODUCT EXPERIENCE

Value by role

Each role needs a compact, repeatable home screen. The product should feel like focused workspaces backed by one domain model, not one giant menu.

Owner / principal

School health, adoption, exceptions, academic trends, billing and governance.

School admin

Tenant setup, academic structure, roster, permissions, reports and support.

Coordinator

Class coverage, teacher completion, assessments, moderation and interventions.

Teacher

Today's classes, attendance, assignments, grading queue and student follow-up.

Student

Today's timeline, due work, attendance, results, recordings and practice arena.

Parent

Authorized child summary, absences, due work, results and school communication.

Access rule

Menus are not security. Every backend action checks membership, tenant, permission and resource ownership. Screens then expose only the tasks relevant to the role.

05 / INITIAL DIRECTION

Ideal customer profile

Concentration beats breadth in the first year. Choose schools with real pain, a reachable buyer and enough budget to implement properly.

Recommended beachhead

India-first private schools in one geographic cluster, roughly 500-2,500 students, digitally active, with a principal or owner who can approve a paid pilot and designate an internal champion.

Qualify in

Spreadsheet or WhatsApp fragmentation; 30+ teachers; reliable internet; named champion; willingness to train; decision inside 60 days; clear renewal owner; need for school-wide reporting.

Qualify out for now

Free-only requirement; custom source ownership; tender-first government sale; offline-only campus; no implementation owner; demand for payroll, transport and accounting before learning workflows.

First wedge

Attendance + assignments + grading + student timeline + management adoption dashboard. Import the roster and integrate or coexist with the existing ERP.

Positioning

The school learning-operations layer: easier to adopt than enterprise LMS products and deeper in academic workflow than broad ERP products.

Initial focus question

- Can one school be live in seven days, can 70% of teachers use the core loop weekly by day 30, and will the buyer pay to renew after 90 days?

06 / MARKET

Bottom-up India opportunity

Use an auditable school-count model rather than a headline market-research forecast. The values below are scenarios, not revenue forecasts.

1.47M
ALL SCHOOLS
UDISE+ 2024-25 total: 1,471,473.

339,583
PRIVATE UNAIDED
Recognized private unaided schools.

95.9M
PRIVATE ENROLMENT
Students in private unaided schools.

LAYER	SCHOOL COUNT	ANNUAL PRICE ASSUMPTION	IMPLIED ANNUAL OPPORTUNITY
TAM reference	1,471,473	INR 180,000	INR 26,486 crore
SAM reference	339,583	INR 180,000	INR 6,112 crore
Beachhead scenario	25,000 assumed	INR 180,000	INR 450 crore
Early SOM milestone	500 customers	INR 180,000	INR 9 crore ARR

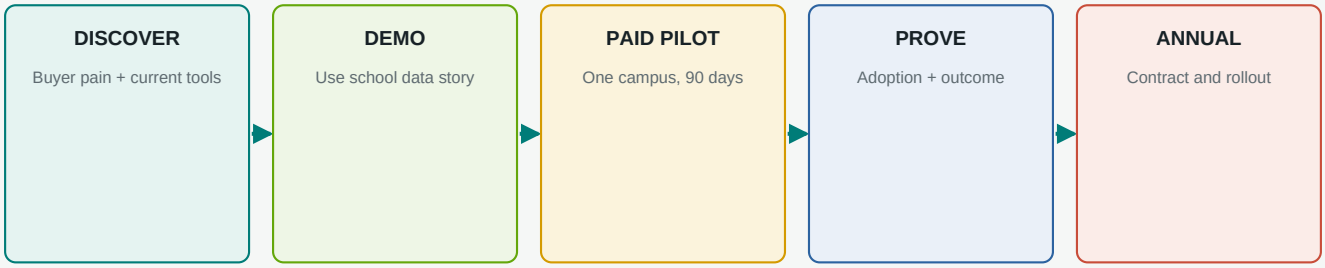
- The SAM is not automatically serviceable. Filter by school size, board, language, region, digital readiness, buyer access and willingness to pay.
- Validate price and reachable school counts with 20-30 buyer interviews. Do not use the TAM slide as a sales forecast.

Source: Ministry of Education, UDISE+ Report 2024-25. Market values are calculated scenarios.

07 / REVENUE

How schools buy

This is a founder-led, organization sale with implementation risk. The purchase is easier when the pilot has explicit success criteria.



ROLE	PRIMARY CONCERN	EVIDENCE REQUIRED
Owner / principal	Control, outcome, reputation, cost	Executive dashboard, success plan, references
Academic head	Teacher consistency and interventions	Workflow demo, reports, moderation
IT / admin	Security, setup, identity, exports	Architecture, DPA, import test, support model
Teacher champion	Speed and usability	Hands-on pilot with real classes
Finance / trust	Contract, payment, liability	Quote, SLA, invoice, data terms

Pilot contract

A pilot is paid professional evaluation, not an indefinite free trial. Define campus, users, modules, migration, training, support, start/end dates, success measures and conversion price.

08 / COMPETITION

Competitive landscape

The market contains strong LMS, classroom and ERP products. Win a narrow purchasing decision instead of claiming to replace every category.

COMPETITOR / CATEGORY	STRONG POSITION	OPENING FOR THIS PRODUCT
Google Classroom	Low-friction classroom workflow; qualifying institutions can use Fundamentals at no charge.	School-specific workflow, implementation, configurable governance and unified operational evidence.
Moodle	Open, flexible, mature and highly customizable.	Managed SaaS, opinionated onboarding, focused role UX and support.
Canvas	Mature K-12 learning, assessment, integrations and analytics ecosystem.	India-first implementation, local workflow, simpler scope and commercial model.
Schoology / PowerSchool	K-12 ecosystem and SIS/LMS connections.	Faster deployment for independent schools and regional organizations.
Teachmint	Connected classroom, AI and classroom hardware offering.	Vendor-neutral software, configuration depth and auditable school operations.
Fedena / school ERP	Broad administration modules including fees, HR, transport and records.	Deeper daily learning loop; integrate rather than rebuild the whole ERP initially.

Competitor descriptions are based on official product pages; see sources.

09 / STRATEGY

How to win without feature warfare

A startup will not outspend platform incumbents. It can win on implementation speed, daily adoption and school-specific operational clarity.

1. Own onboarding

Roster templates, validation, migration preview, admin checklist, teacher training and a seven-day go-live target.

2. Make teachers faster

Attendance in under two minutes, assignment reuse, bulk grading, saved rubrics, mobile-responsive queues and reliable notifications.

3. Give management evidence

Adoption, completion, exceptions, interventions and trend reports that answer operational questions without exports.

4. Configure, do not fork

Settings, entitlements, templates and adapters cover valid variation. No tenant-specific branches in core code.

5. Integrate pragmatically

CSV first, then OneRoster, Google/Microsoft identity, payment, messaging, video and selected ERP connectors.

6. Build trust

Tenant isolation, audited changes, clear data terms, exportability, backups, incident readiness and responsive support.

Defensible advantage

The moat is accumulated implementation knowledge, tenant configuration, workflow adoption data, integration reliability and customer references - not a large list of commodity LMS features.

10 / PRODUCT

MVP scope

Build only enough to run and measure the complete school learning loop for real pilot campuses.

DOMAIN	MVP CAPABILITY	ACCEPTANCE TEST
Tenant + identity	Organizations, memberships, roles, login, invitations, support access	No cross-tenant access; role tests pass
Academic setup	Year, terms, grades/programs, classes, subjects, teacher allocation	Roster imports with validation report
Class operations	Sessions, attendance, timeline, announcements	Teacher completes attendance quickly
Assignments	Create, schedule, attach, submit, comment, grade, rubric	Student-to-grade loop works end to end
Assessments / Arena	Question bank, practice sets, attempts, scoring, teacher review	Repeatable practice and result history
Student / parent view	Due work, attendance, marks, recordings, notifications	Only authorized child data is visible
Management	Adoption, completion, exception and tenant usage dashboard	Principal answers weekly questions
Platform operations	Plans, entitlements, billing status, audit, usage and support	Tenant can be provisioned and suspended

11 / SCOPE CONTROL

What not to build first

A full school ERP is a separate company-sized scope. Preserve the option to integrate before attempting replacement.

Defer ERP breadth

Fees, payroll, accounting, transport, hostel, library, inventory, admissions CRM and biometric hardware.

Defer platform complexity

Microservices, Kubernetes, multi-region active-active, event sourcing, custom DSL and generalized workflow engines.

Defer data stores

MongoDB and search clusters until PostgreSQL JSONB and indexed full text become measured bottlenecks.

Defer AI autonomy

No automatic publishing, final grading, disciplinary decisions or student risk labels without a human owner.

Do integrate

Identity, roster import, video provider, email/SMS/WhatsApp, payment and exports needed by paid pilots.

Do preserve exits

Versioned APIs, event outbox, stable IDs and clean module ownership make later extraction possible.

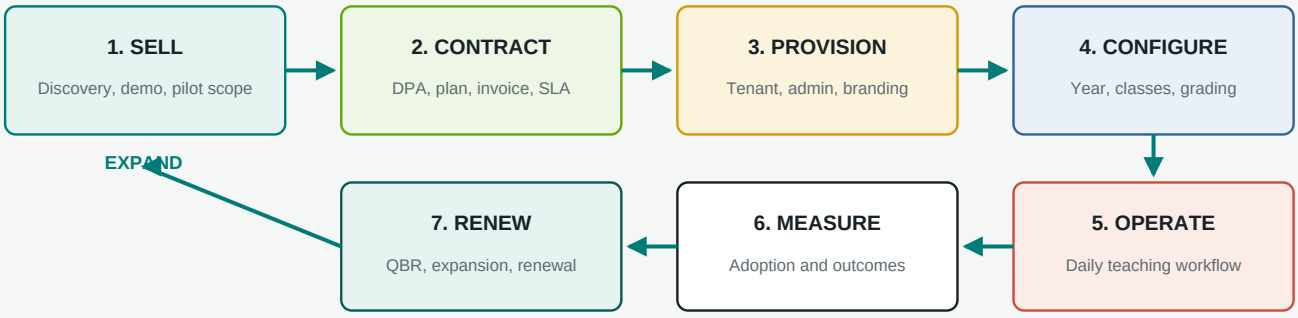
Scope test

A requested feature enters the core only if it repeats across target schools, supports the chosen learning loop, improves sales or renewal, and can be operated securely.

12 / FULL FLOW

Product lifecycle: sale to renewal

The product flow begins before login and ends with renewal. Implementation, support and measurement are part of the service.

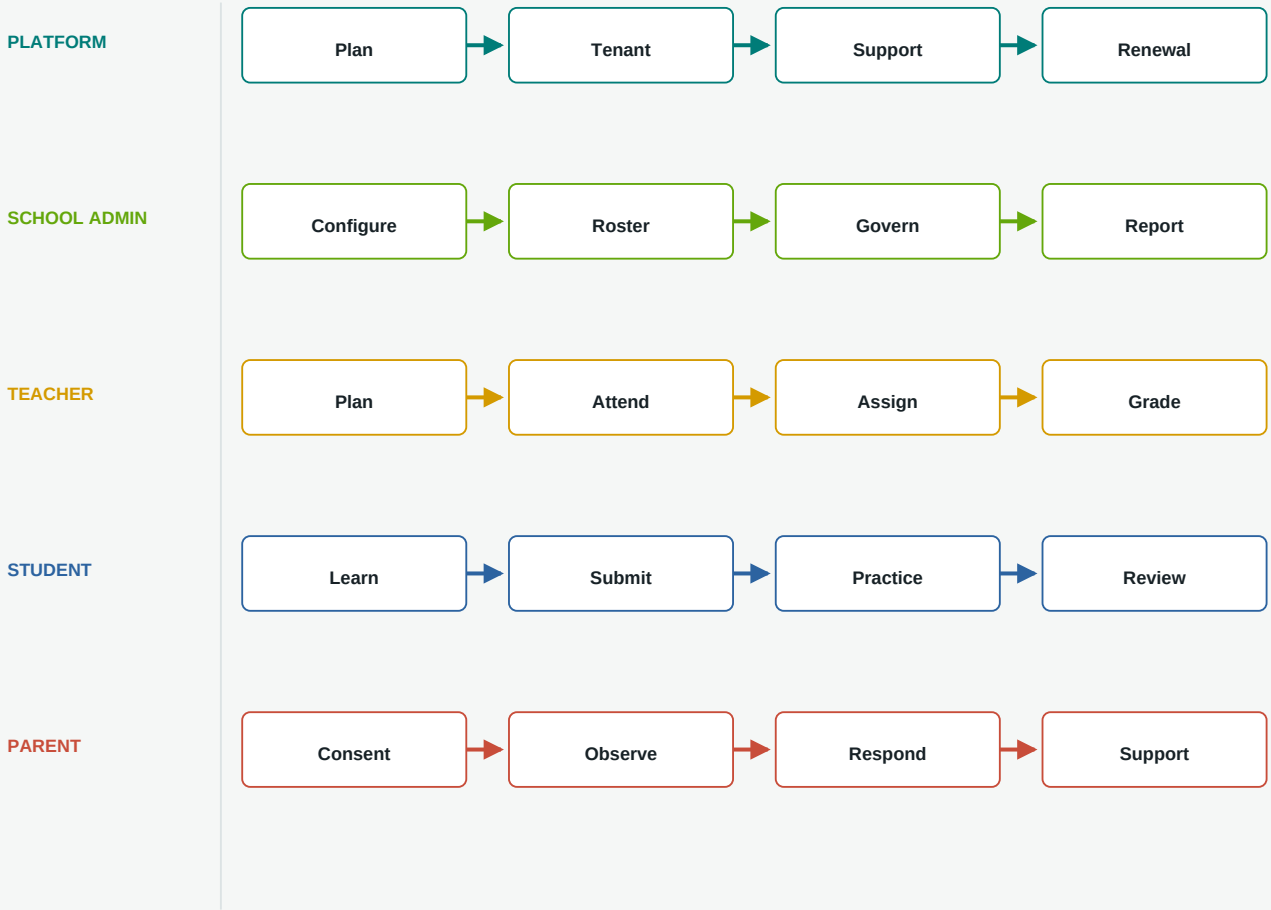


- Sales hands a structured implementation record to customer success; it does not hand over meeting notes.
- Provisioning creates the tenant and first admin, then configuration and roster import make the tenant operational.
- Daily usage emits product events and domain records. Dashboards summarize both learning workflow and adoption.
- Quarterly business reviews compare agreed success metrics, risks, support history and expansion opportunities.

13 / FULL FLOW

Role-by-role operating lanes

The platform is one coordinated system, but responsibility remains explicit across platform staff, school staff, learners and families.



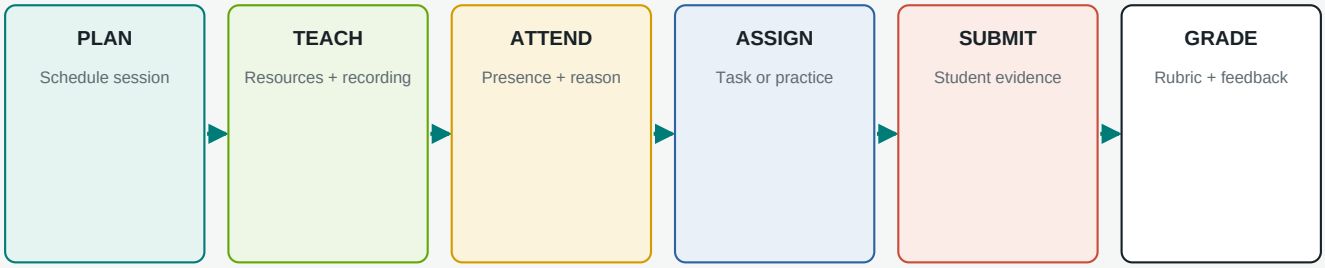
Key handoffs

The platform provisions; the school configures; teachers create authoritative learning records; students respond; parents observe within consent and policy; management acts on exceptions.

14 / LEARNING LOOP

Daily classroom workflow

This loop should be the fastest and most reliable part of the product because it creates the data that powers every dashboard.



Teacher experience

Default to today's classes; reuse content; support bulk actions; autosave drafts; show exactly what will be published; make corrections audited.

Student experience

One ordered timeline; clear due dates and status; mobile-first submission; low-bandwidth access; transparent feedback and retry rules.

Management experience

Track missing attendance, unpublished grades, overdue work, low completion and intervention queues without opening individual classes.

Notification behavior

Domain events create scheduled, deduplicated notifications according to school policy, quiet hours, channel and recipient consent.

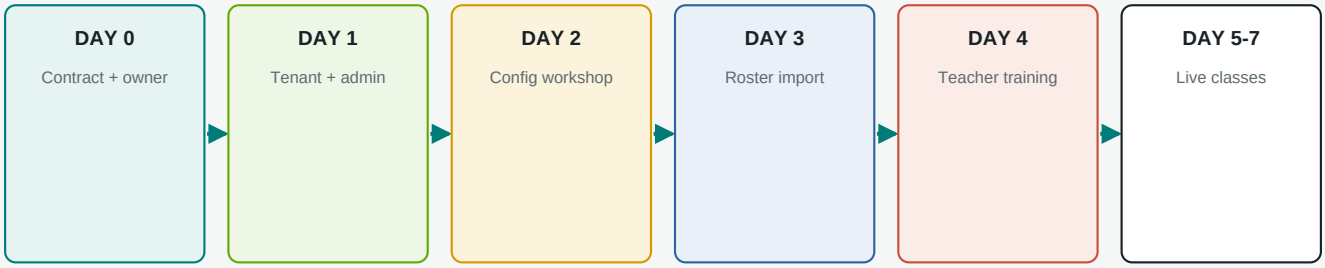
Primary product event chain

ClassSessionStarted -> AttendanceRecorded -> AssignmentPublished -> SubmissionReceived -> GradeReleased -> InterventionOpened -> InterventionResolved

15 / IMPLEMENTATION

Seven-day tenant onboarding

Fast time-to-value is a product capability. Make imports, validation and training repeatable before adding more modules.



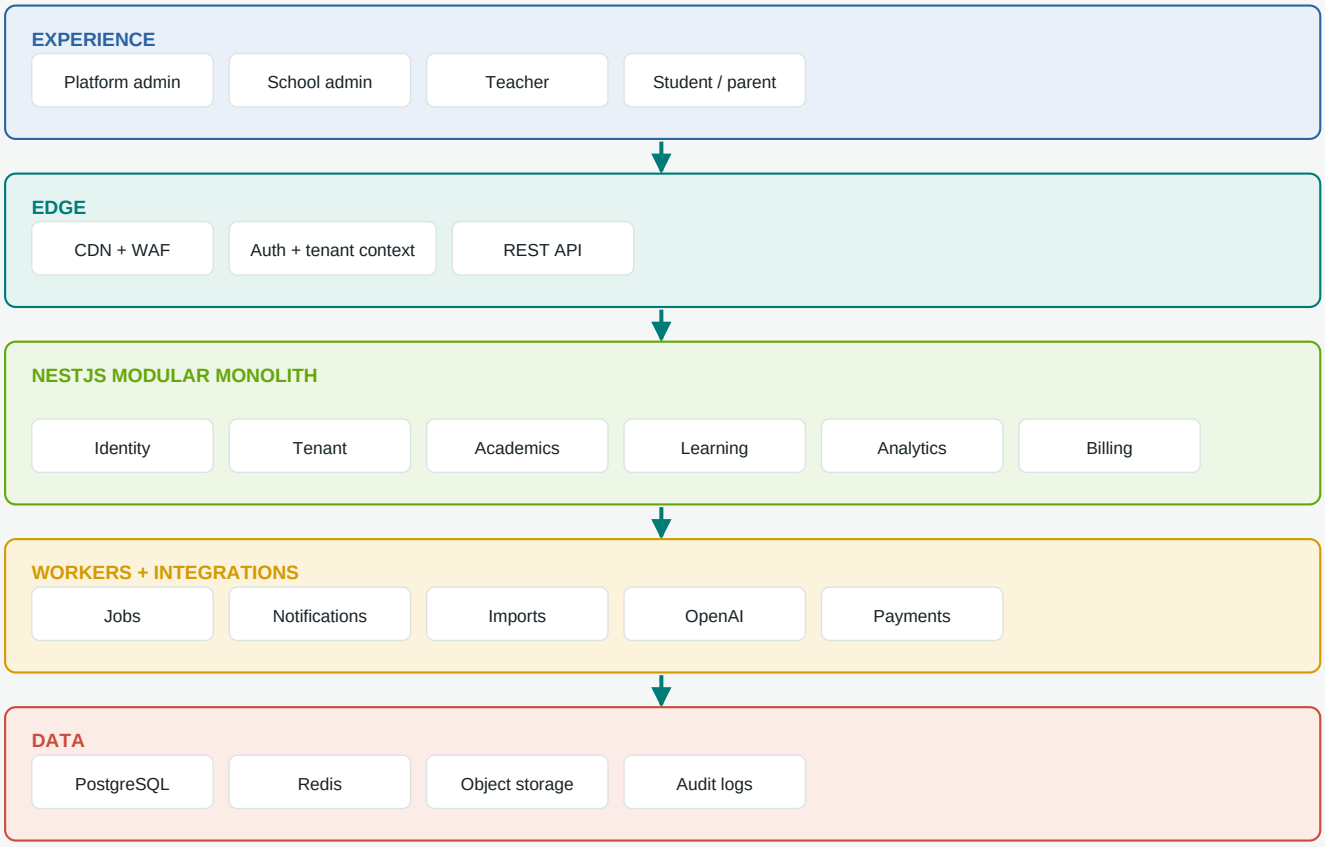
WORKSTREAM	OWNER	DEFINITION OF DONE
Commercial	Sales / CS	Signed order, DPA, plan, invoice, success metrics
Configuration	School admin + CS	Branding, year, terms, grading, attendance and channels approved
Data	School admin	Students, staff, classes and enrollments pass validation
Training	Teacher champion	Champion and pilot teachers complete real workflow
Go-live	CS	First attendance and assignment recorded
Adoption	CS + principal	Weekly review and corrective actions active

- Import preview must show duplicates, missing identifiers, invalid classes and proposed changes before commit.
- Every onboarding step is represented in the product; avoid running the entire process from private spreadsheets.

16 / ARCHITECTURE

Recommended system architecture

Use the proposed TypeScript stack, but deploy the backend as a modular monolith first. The internal boundaries remain explicit without paying the distributed-system cost.



LAYER	RECOMMENDATION	REASON
Web	Next.js + TypeScript	One role-aware application; PWA/mobile later
API	NestJS REST + OpenAPI	Clear modules, validation, guards and generated clients
Database	PostgreSQL	Transactions, reporting, JSONB and tenant-scoped records
Async / cache	Redis + BullMQ	Jobs, rate limits and ephemeral cache; never system of record
Files / video	S3-compatible storage + video provider	Direct uploads and signed access; do not proxy large files
AI	OpenAI from NestJS worker	No FastAPI until a distinct Python/ML workload exists

17 / BACKEND

Modular monolith boundaries

Each module owns behavior and tables, exposes application services, and publishes domain events through an outbox. Avoid direct cross-module table edits.

MODULE	OWNS	PUBLISHES / CONSUMES
Identity	Accounts, credentials, MFA, sessions	UserActivated, LoginRiskDetected
Tenant	Organizations, campuses, membership, settings	TenantProvisioned, MembershipChanged
Entitlements	Plans, limits, feature access	PlanChanged, LimitApproaching
Academics	Years, terms, programs, classes, subjects, enrollment	ClassCreated, EnrollmentChanged
Class operations	Sessions, attendance, announcements	AttendanceRecorded, ClassClosed
Learning	Assignments, submissions, questions, attempts, grades	AssignmentPublished, GradeReleased
Content	Files, resources, recordings, access policy	AssetReady, RecordingPublished
Communication	Templates, preferences, delivery history	Consumes domain events
Analytics	Projections, metrics and exports	Consumes all relevant events
Billing	Subscription, invoice state, usage ledger	SubscriptionChanged, PaymentFailed
Audit	Immutable security and business activity	Consumes privileged actions
Integrations	Imports, mappings, webhooks and provider adapters	SyncCompleted, SyncFailed

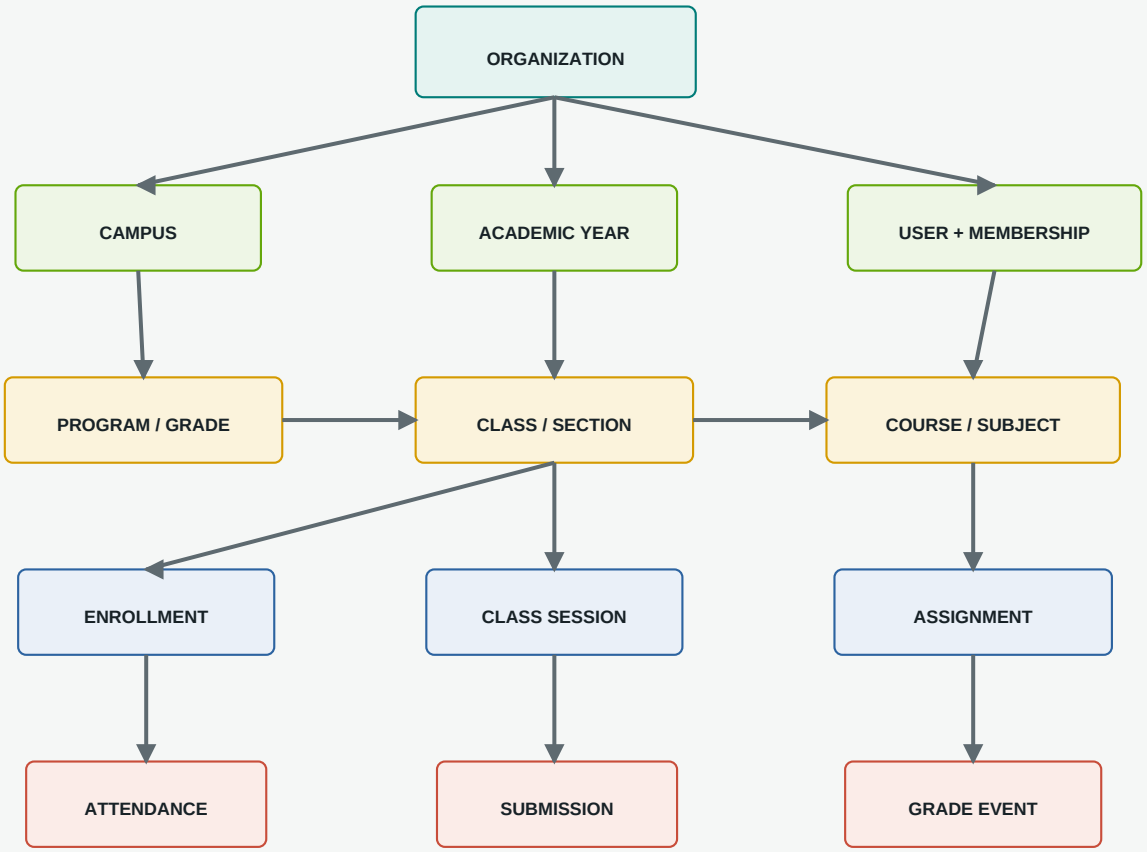
Deployment shape

Run the same codebase as an API process and a worker process. Extract a module only when independent scaling, ownership, reliability or deployment evidence justifies it.

18 / DATA

Core organization data model

Tenant identity sits above the academic hierarchy. Operational records inherit tenant scope through their authoritative parent and also carry tenant_id for enforcement and indexing.

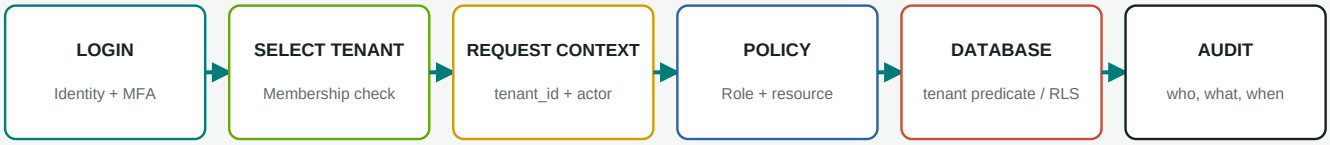


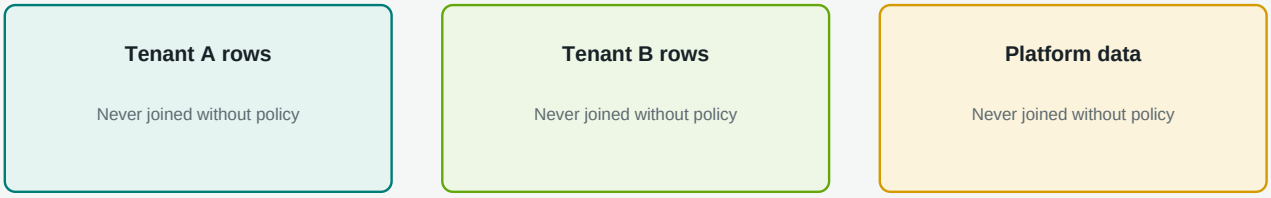
- Use immutable public IDs; keep internal numeric or UUID keys private to the service.
- Model user identity separately from organization membership so one person can hold different roles in different tenants.
- Represent grades as append-only grade events plus current projection when auditability matters.
- Store configuration and question variants in validated PostgreSQL JSONB; move only when usage proves a different store is necessary.

19 / TENANCY

Tenant isolation end to end

A tenant_id column is necessary but not sufficient. The boundary must survive authentication, policy, SQL, cache, jobs, files, analytics, exports and support tooling.





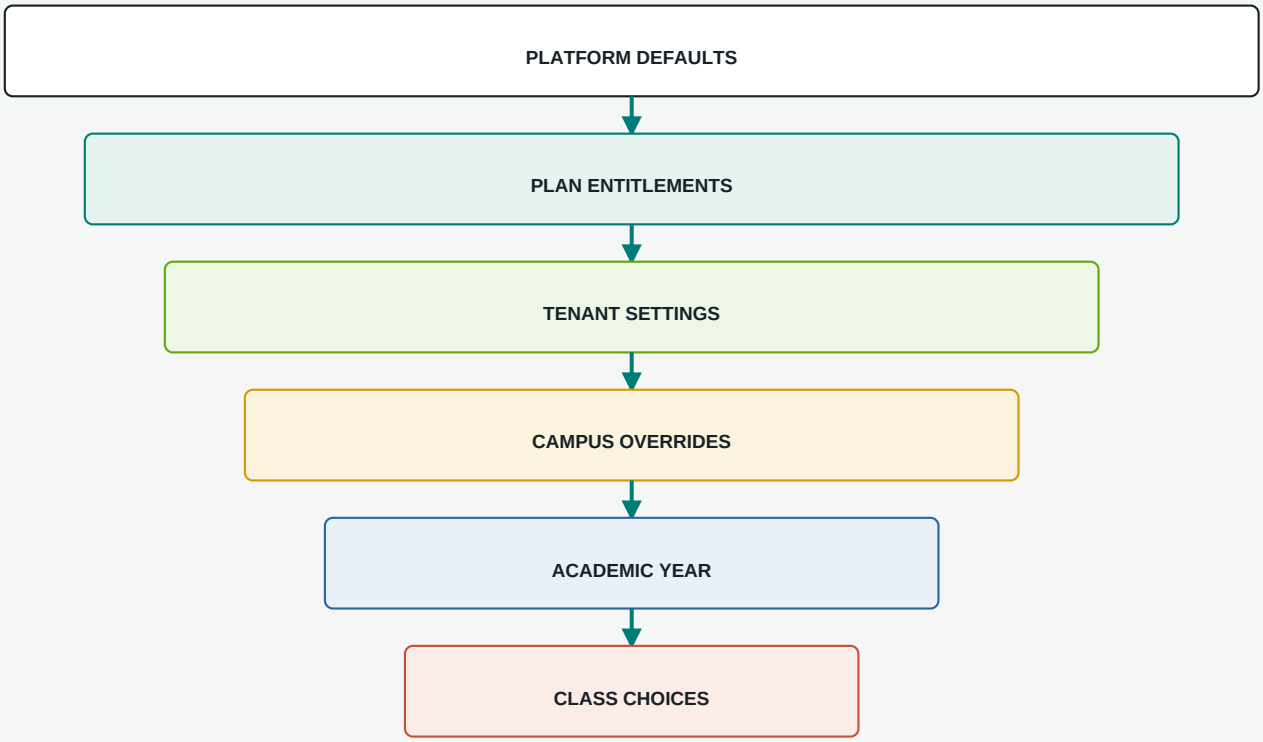
Backups, exports, files, cache keys and jobs carry the same tenant boundary.

CONTROL	IMPLEMENTATION
Request context	Resolve tenant from selected membership; never trust a raw client tenant_id
Authorization	Policy checks use actor, tenant, action and resource; deny by default
Database	Tenant predicate in every query; consider PostgreSQL RLS as defense in depth
Uniqueness	Unique constraints include tenant_id where values are tenant-local
Cache / jobs	Prefix keys and job payloads with tenant; workers re-authorize sensitive work
Files	Tenant path prefix, signed URLs, metadata check and short expiration
Support	Just-in-time, reason-bound impersonation with prominent banner and full audit
Tests	Automated cross-tenant negative tests for every major resource type

20 / CONFIGURATION

Configuration hierarchy

Resolve settings predictably. Lower levels may override only keys that the configuration schema explicitly allows.



- Resolution order: platform defaults -> plan entitlements -> tenant -> campus -> academic year -> class.
- Every setting has a typed schema, default, allowed scope, validation rule, version and audit history.
- Publish configuration changes through a draft/review/activate flow when they affect grades, attendance or access.
- Snapshot high-impact settings onto published assessments or reports so later changes do not rewrite history.

Do not build a generic rules engine

Start with explicit settings and small state machines. Add workflow composition only after repeated customer patterns are understood.

21 / CONFIGURATION

Dynamic without breaking the core

The product should adapt to valid school variation while keeping security, accounting and academic history consistent.

CONFIGURABLE BY TENANT	NEVER TENANT-CUSTOMIZED IN CORE CODE
Brand, domain, locale, time zone, terminology	Tenant isolation and authorization checks
Academic year, terms, grade/program hierarchy	Audit event identity and timestamp
Attendance statuses, late rules and correction approvals	Published record integrity and ownership
Grading scales, weights, rubrics and report layouts	Subscription enforcement and usage accounting
Assignment types, practice settings and release policy	Encryption, backups and incident logging
Notification channels, templates, quiet hours and escalation	API versioning and idempotency behavior
Role templates and delegated administration	Child safety controls and privileged support audit
Enabled modules, storage and AI budgets	Core entity identifiers and referential integrity

Configuration storage

- tenant_settings(tenant_id, key, schema_version, value_jsonb, status, effective_at, updated_by).
- feature_entitlements(tenant_id, feature_key, enabled, limit_value, source_plan, effective_window).
- configuration_audit(before, after, reason, actor, approval, timestamp).
- Use JSON Schema or equivalent runtime validation. Reject unknown keys; migrate versions explicitly.

22 / INTEGRATION

APIs, jobs and ecosystem

Schools already have identity, spreadsheets, communication channels and sometimes ERP/SIS products. Integration is a sales feature and an operating risk.

Synchronous API

REST + OpenAPI, versioned endpoints, pagination, idempotency keys for writes, clear errors and tenant-aware rate limits.

Asynchronous work

Transactional outbox -> BullMQ workers for notifications, imports, exports, analytics projections, AI and provider retries.

Import / export

CSV templates first with preview and reconciliation. Preserve stable external IDs and downloadable tenant exports.

School standards

Adopt OneRoster 1.2 for roster and grade exchange when integrations justify it; add LTI 1.3 / LTI Advantage for external learning tools later.

Webhooks

Signed delivery, event IDs, retry policy, dead-letter review and per-tenant secrets. Never include unnecessary child data.

Provider adapters

Identity, payments, email, SMS/WhatsApp, video and storage sit behind small interfaces with usage and failure telemetry.

Integration order

1) CSV roster, 2) Google/Microsoft sign-in, 3) messaging and payment, 4) selected ERP/SIS connector, 5) OneRoster/LTI certification when enterprise demand is real.

23 / AI

Responsible AI product strategy

AI should reduce teacher preparation and review time. It must not become an unreviewed authority over children or academic records.

USE CASE	HUMAN CONTROL	DATA RULE
Draft practice questions	Teacher reviews, edits and publishes	Prefer curriculum context; omit student identity
Rubric / feedback draft	Teacher owns final feedback and grade	Send minimum submission content
Summarize class activity	Staff validates before sharing	Tenant-scoped sources and permissions
Translate communication	School approves templates	Do not infer sensitive attributes
Student study assistant	Bounded to approved content and age policy	Log safety events; no behavioral profiling
Risk or intervention suggestions	Explainable signal only; staff decides	No automated adverse decision

- Call OpenAI from a NestJS worker first. Add FastAPI only when a Python-specific model or pipeline has independent deployment needs.
- Create an AI gateway module for prompt versions, model allowlist, redaction, usage budgets, structured outputs, evaluation and audit.
- Default to store=false where supported; verify provider retention and residency terms for the chosen endpoints and contract.
- Measure acceptance rate, teacher time saved, unsafe-output rate, cost per task and override reasons.

OpenAI API data-control details change over time; verify current official documentation before production.

24 / TRUST

Security, privacy and child data

Security is a procurement requirement and a product behavior. Build evidence continuously instead of assembling it during a school review.

Identity

OIDC, MFA for privileged users, secure password policy, session revocation, device/session view and short-lived support access.

Authorization

Deny-by-default policies, tenant/resource checks, least privilege, delegated admin and periodic role review.

Data protection

TLS, encryption at rest, secrets manager, least-data collection, retention schedules, exports and deletion workflow.

Application security

Dependency scanning, SAST, secure headers, rate limits, upload scanning, signed URLs and penetration testing before scale.

Resilience

Automated backups, point-in-time recovery, restore drills, incident runbooks, status communication and provider fallback.

Evidence

Audit logs, subprocessors, DPA, access review, restore evidence, vulnerability process and customer-facing security pack.

India compliance note

The DPDP Act and notified DPDP Rules include duties covering safeguards, breach handling, retention and children's data. Educational exemptions are purpose- and condition-specific. Obtain Indian privacy counsel before launch and track phased commencement dates.

25 / RELIABILITY

Non-functional requirements

Write measurable service objectives before school-wide rollout. Reliability targets shape architecture, support promises and pricing.

AREA	INITIAL OBJECTIVE	EVIDENCE
Availability	99.9% monthly for core API excluding announced maintenance	Synthetic checks and monthly report
Performance	p95 API reads under 500 ms; attendance write under 800 ms	Tenant-aware tracing and dashboards
Recovery	RPO <= 15 minutes; RTO <= 4 hours for core records	Quarterly restore drill
Durability	Direct object storage and checksummed uploads	Provider durability + verification
Capacity	Load test at 3x largest contracted concurrent usage	Pre-term capacity review
Support	P1 acknowledge 30 minutes; P2 four business hours	Ticket timestamps and incident review
Observability	Trace request -> SQL/job/provider with tenant-safe metadata	Logs, metrics and traces
Change safety	Backward-compatible migrations and feature-controlled rollout	CI checks and rollback drill

- Never put student names, submissions, tokens or sensitive payloads in normal application logs.
- Measure queue delay, provider failure, import error rate, notification delivery, slow queries and cross-tenant denial events.

26 / EXECUTION

First 90 days

Do discovery and implementation in parallel, but refuse feature commitments that are not tied to the target workflow or pilot success.

PERIOD	PRODUCT / ENGINEERING	COMMERCIAL / CUSTOMER	EXIT GATE
Days 1-15	Domain model, tenant boundary, clickable workflow, threat model	20 buyer/teacher interviews; shortlist 5 pilot schools	Three paid-pilot candidates
Days 16-30	Tenant, identity, academics, roster import, audit foundation	Pilot proposal, contract, onboarding checklist	One signed paid pilot
Days 31-60	Sessions, attendance, assignments, submissions, grading, notifications	Configure and train first campus; daily support	Real classes complete core loop
Days 61-75	Student timeline, management adoption dashboard, exports	Weekly success review and usability fixes	70% pilot teachers weekly active
Days 76-90	Reliability, security pack, billing, renewal workflow	Outcome review, reference and annual conversion	At least one annual conversion

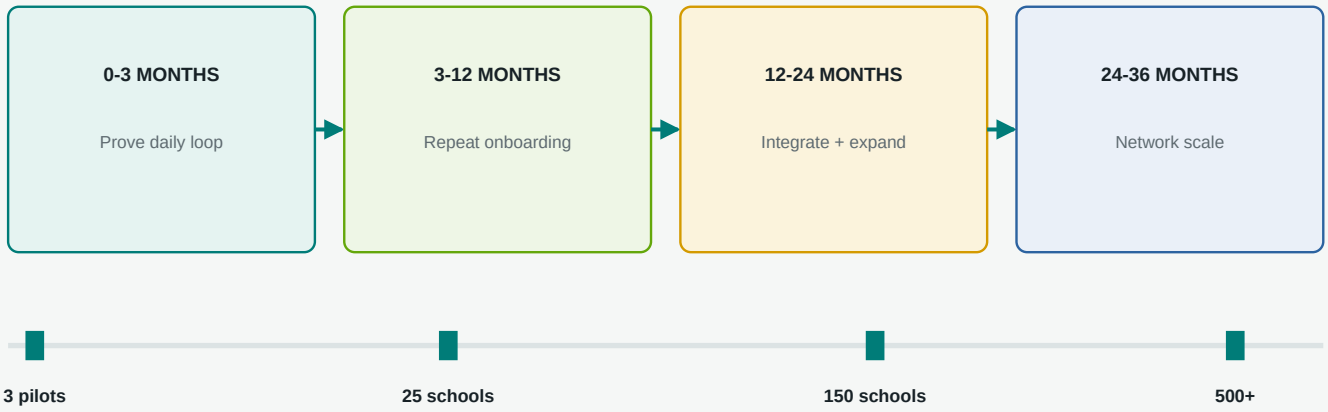
The 90-day rule

Do not begin the next major module until a real school has completed the current workflow, usage has been measured, and the buyer can explain its value.

27 / GROWTH

12-36 month roadmap

Expand by making deployment repeatable, improving retention and adding integrations. Do not translate early customer requests directly into permanent architecture.



PHASE	BUSINESS	PRODUCT / PLATFORM
0-3 months	3 paid pilots; prove conversion	Core academic loop, onboarding, isolation, audit
3-12 months	25-50 schools in one region; references and partner channel tests	Parent view, better analytics, templates, support tooling, first integrations
12-24 months	150+ schools; second region; school groups	Multi-campus governance, OneRoster, LTI, mobile app only if usage supports it
24-36 months	500+ schools; repeatable channels and expansion	Enterprise SSO, data warehouse, regionalization, selective service extraction

- Possible later adjacencies: admissions integration, fee visibility, content partnerships, assessment network and benchmarking - only with consent, privacy and buyer demand.

28 / PRICING

Monthly price per organization

Quote a monthly equivalent for clarity, but contract and collect annually. Charge onboarding separately because migration and training create real cost.

PLAN	MONTHLY EQUIVALENT	INCLUDED	ONE-TIME ONBOARDING
Paid pilot	INR 30,000 / 90 days	1 campus, up to 800 students, core workflow, training, success review	Included
Foundation	INR 8,000 / month	Up to 500 active students, core modules, email support	INR 20,000
Growth	INR 15,000 / month	Up to 1,500 students, analytics, parent access, priority support	INR 40,000
Scale	INR 30,000 / month	Up to 4,000 students, advanced controls, integrations allowance	INR 75,000
School group	Custom	Group reporting, multiple campuses, SSO, SLA and integration scope	Scoped

Add-ons and limits

- Pass-through usage plus margin for SMS/WhatsApp, premium video, additional storage, AI credits and payment fees.
- Charge separately for complex historical migration, custom reports, private training days, dedicated environments and non-standard integrations.
- Do not discount recurring price to hide implementation cost. Discount only for annual prepayment, reference rights or multi-campus volume.
- Validate willingness to pay through signed pilots. Pricing is a starting hypothesis, not a promise.

29 / ECONOMICS

Unit economics and operating metrics

Revenue quality matters more than tenant count. A low-priced school that requires permanent customization and support can destroy margin.

>75%
GROSS MARGIN TARGET
After cloud, providers and direct support.

<90 days
CAC PAYBACK GOAL
After the sales motion becomes repeatable.

>110%
NET REVENUE RETENTION
From upgrades and campus expansion.

METRIC GROUP	TRACK WEEKLY / MONTHLY
Acquisition	Qualified schools, demo-to-pilot, pilot-to-paid, sales cycle, CAC
Activation	Days to tenant live, roster error rate, teachers trained, first class event
Adoption	Weekly active teachers, attendance completion, assignment publish and grading completion
Learning operations	Student completion, feedback time, intervention open/resolve, parent reach
Retention	Logo churn, gross retention, expansion, support health, renewal forecast
Economics	ARR, ARPA, gross margin, cloud cost/tenant, provider cost/student, support hours/tenant
Reliability	Availability, latency, incidents, restore results, notification and job failure

Leading renewal indicator

Weekly active teacher percentage combined with completion of the agreed core workflow is more useful than total logins.

30 / RISK

Challenges and blockers

The largest risks are commercial and behavioral, not only technical. Give each one an owner and an early warning signal.

RISK	EARLY SIGNAL	MITIGATION
Customization pressure	Roadmap dominated by one tenant	Configuration policy, paid integration scope, product council
Teacher resistance	Low weekly workflow completion	Champion, training, fast UX, principal follow-up
Bad source data	Import errors and duplicates	Preview, reconciliation, stable external IDs
Long sales / collections	Unsigned pilots, overdue invoices	Qualify authority, paid pilot, annual prepay
Scope expansion to ERP	Every deal requires unrelated modules	Choose wedge; integrate existing ERP
Child-data incident	Overbroad access or logging	Least data, reviews, audit, incident drills, counsel
AI error or cost	Low acceptance, high spend	Human approval, evals, budgets, model routing
Tenant leakage	Missing tenant filters in tests	Central context, RLS defense, negative tests
Support overload	Hours/tenant rise with growth	In-product onboarding, runbooks, telemetry, plan boundaries
Vendor dependency	Provider failure blocks core flow	Adapters, queues, fallback and exportability

31 / TEAM

Initial team and operating model

A small team can ship the first version if it owns complete outcomes and avoids distributed architecture.

RESPONSIBILITY	INITIAL OWNER	PRIMARY OUTPUT
Product + founder sales	Founder / product lead	ICP, discovery, roadmap, pilot and pricing
Backend / platform	2 engineers	NestJS modules, data, tenancy, integrations and reliability
Frontend / experience	1-2 engineers	Role workspaces, responsive workflow and accessibility
Design / research	Shared or fractional	Teacher workflow, usability tests and design system
Customer success	1 implementation lead	Configuration, imports, training, adoption and renewal
Security / privacy	Named engineering owner + counsel	Controls, DPA, review, incident readiness
QA / automation	Team-owned initially	Critical-path, tenant-isolation and regression tests

Operating cadence

- Weekly: customer usage review, pilot blockers, reliability and support trends.
- Biweekly: ship a thin end-to-end increment, test with teachers, and review security impact.
- Monthly: product council across recurring requests; pricing and unit-economics review.
- Quarterly: customer outcome review, restore exercise, access review and roadmap reset.

32 / DECISION MEMO

What to do next

The company should prove a repeatable school outcome before expanding the platform surface.

Recommended decision

Proceed with a 90-day paid-pilot program for the learning-operations wedge, backed by a modular multi-tenant architecture and a strict no-fork configuration policy.

PRIORITY	DECISION
1	Choose one city/region and one school profile; conduct 20 buyer and teacher interviews.
2	Secure three paid-pilot candidates before committing to broad build scope.
3	Implement tenant, identity, roster, class, attendance, assignment, grading, timeline and management adoption.
4	Define security, child-data, retention, support and contract responsibilities before production data.
5	Launch one campus in seven days and measure teacher adoption weekly.
6	Convert at least one pilot to annual payment before adding major ERP or AI scope.
7	Use repeatable configuration and integration patterns; reject tenant-specific core branches.
8	Revisit microservices, MongoDB, FastAPI and mobile native apps only with measured need.

Success at day 90

A paying school runs real classes, most pilot teachers complete the weekly workflow, management acts on the dashboard, support is controlled, and the buyer signs or commits to annual renewal.

33 / SOURCES

Primary sources and assumptions

Sources were accessed for this report in September 2026. Product capabilities and laws change; re-check before procurement, contracting or production launch.

Market baseline

Ministry of Education, UDISE+ Report 2024-25, national highlights and school-management tables.
education.gov.in/.../UDISE+Report+2024-25+-+NEP+Structure.pdf

Ministry statistics index.
dsel.education.gov.in/statistics

Competitor product references

Google Classroom editions and organization features.
edu.google.com/workspace-for-education/products/classroom/editions/

Instructure Canvas for K-12.
instructure.com/k12

Moodle for K-12.
moodle.com/solutions/k-12/

Teachmint Connected Classroom platform.
teachmint.com/en-us

Fedena feature tour and product clarification.
fedena.com/feature-tour
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Standards, privacy and AI data controls

1EdTech OneRoster 1.2 and LTI Advantage.
standards.1edtech.org/oneroster/specifications/standards/v1p2
1edtech.org/standards/lti/suggested-lti-advantage-requirements-rfps

India Code, Digital Personal Data Protection Act, 2023; MeitY, DPDP Rules, 2025.
indiacode.nic.in/indiacode/handle/123456789/22037
meity.gov.in/documents/.../digital-personal-data-protection-rules-2025

OpenAI API data controls.
platform.openai.com/docs/models/default-usage-policies-by-endpoint

Important

Market values and prices in this report are planning assumptions, not independent valuations. Security and privacy sections are product guidance, not legal advice.